Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Experienced in developing and evaluating machine learning models, AI-powered applications, and scalable backend services using Python, PyTorch, FastAPI, Docker, and AWS. Hands-on experience in Generative AI, NLP, Computer Vision, model evaluation, and deploying production-ready AI solutions. Published IEEE researcher with experience designing experimental pipelines, benchmarking model performance, and translating research into practical applications.

SKILLS

- **Programming :** Python, Java, SQL, C++, Bash
- **Machine Learning & AI :** PyTorch, TensorFlow, Keras, Scikit-Learn, Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), NLP, Computer Vision, Vision Transformers, Continual Learning, Transfer Learning, Hugging Face Transformers, Model Training, Model Evaluation, Inference Pipelines, Hyperparameter Tuning
- **Data Science & Analytics :** Pandas, NumPy, Matplotlib, Data Analysis, Statistical Analysis, Feature Engineering, Data Preprocessing, Statistical Modeling
- **Backend & APIs :** FastAPI, REST APIs
- **Model Risk & Analytics :** Model Validation, Performance Benchmarking, Predictive Modelling, Statistical Analysis, Reliability Assessment
- **Tools :** Git, GitHub, Linux, Docker, AWS

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Computer Vision, Risk Management, Deep Learning and Applied AI, Natural Language Processing, Machine Learning, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) **Nov 2025**

- Conducted systematic evaluation of pretrained Vision Transformer architectures under continual learning settings, analyzing model robustness, catastrophic forgetting, and knowledge retention across sequential tasks.
- Designed experimental pipelines for training, evaluation, and comparison of computer vision models across sequential tasks.
- Evaluated model performance under changing data distributions and analyzed model reliability.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Developed reproducible training and evaluation pipelines using Python and PyTorch.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] **May 2023**

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Evaluated supervised ML algorithms including Random Forest, SVM, Decision Trees, and Neural Networks across environmental datasets.
- Performed comparative analysis using model performance metrics to assess accuracy, robustness, and generalization.
- Authored and presented findings through an IEEE peer-reviewed publication.

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an AI-powered document intelligence application with backend APIs for automated document analysis and semantic comparison.
- Designed RESTful endpoints using FastAPI for automated document analysis and change tracking workflows.
- Built backend processing pipelines for document ingestion, semantic analysis, and comparison workflows.
- Integrated transformer-based NLP models into production APIs for semantic similarity and document intelligence tasks.
- Developed scalable inference endpoints using FastAPI and containerized deployment workflows using Docker and AWS ECS Fargate.
- Implemented semantic matching to identify added, removed, and modified content across document versions.
- Configured application monitoring and logging using AWS CloudWatch.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated defect detection using CNN, SVM, and Random Forest.
- Built preprocessing, training, and evaluation pipelines.
- Compared deep learning and traditional ML approaches.
- Performed model evaluation using accuracy, confusion matrices, and performance analysis.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer embeddings, quantization, and reconstruction techniques.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Analyzed trade-offs between compression efficiency and semantic preservation across multiple quantization levels.
- **Technologies:** Python, PyTorch, Transformer Models, NLP, Sentence Embeddings